SciFig-Evaluation: A Multilingual Benchmark and Behavioural Framework for Evaluating Multimodal Language Models on Scientific Figures

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Declaration

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

Signed:

Date: 2026

Abstract

Multimodal large language models are increasingly asked to read scientific figures, yet existing benchmarks largely measure whether a model can extract a value from a chart, not whether it is a faithful visual reader. This thesis introduces SCIFIG-EVAL, a multilingual benchmark of 1,005 figures drawn from 158 arXiv papers and paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, together with a three-axis behavioural framework — Admittance, Resistance and Inductance — that separates honesty in the absence of evidence, faithfulness under misleading context, and sound inductive reasoning. Using an MQM-style multi-judge evaluation pipeline, the framework is exercised on eleven contemporary models spanning five families via five adversarial probe types. The resulting behavioural profile reveals systematic, near-orthogonal failure modes that conventional accuracy scores obscure.

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Contents

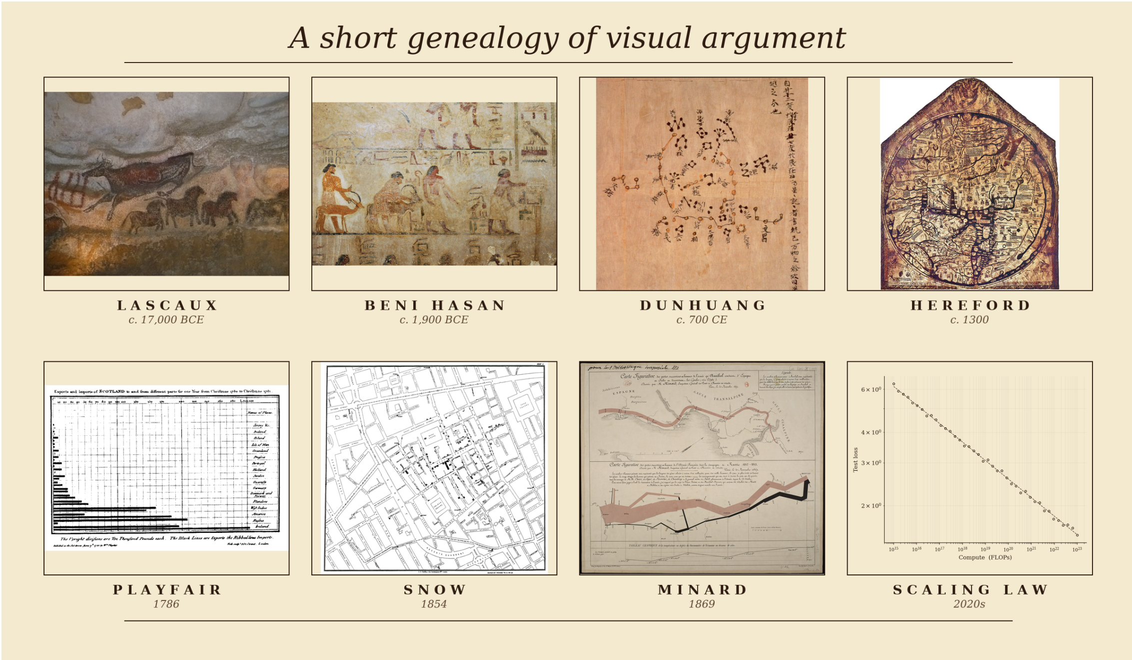
Chapter 1

Introduction

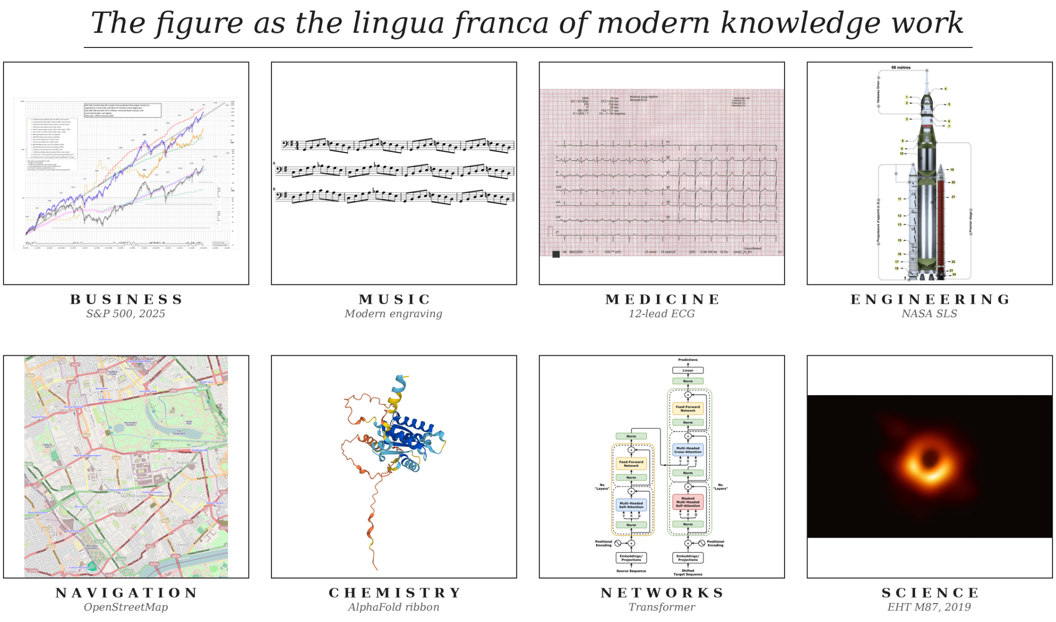
1.1 *Homo sapiens*: A Species of Visual Thinkers

The record of a species willing to reason pictorially begins no later than the Upper Palaeolithic. Some time between seventeen and fifteen millennia before the present, an inhabitant of what is now the Dordogne crushed haematite between stones, lit a pine torch, and rendered an aurochs on Lascaux limestone at approximately life size (Aujoulat, 2005). These visual arguments have continued to evolve. Twelfth-dynasty Egyptian painters at Beni Hasan committed a caravan of Semitic travellers, traditionally associated with the Joseph narrative, to a tomb wall around 1900 BCE (Kamrin, 2009); seventh-century Chinese astronomers committed over fourteen hundred stars to silk at Dunhuang (Bonnet-Bidaud et al., 2009); medieval cartographers compressed theology, geography and chronology onto the Hereford *mappa mundi* (Harvey, 1996). In 1786 William Playfair invented the bar chart in order to argue, graphically and in print, about Britain's trade balances (Playfair, 1786). John Snow's 1854 cholera map of Soho laid down the method of modern epidemiology (Johnson, 2006), and Charles Joseph Minard's 1869 flow map of Napoleon's 1812 Russian campaign (Friendly, 2002) is what Tufte (2001) has called, without serious contest, the densest argument ever made on paper (Figure 1.1a). When a human being intends a claim of any real quantitative weight, the first move is almost invariably to draw.

In contemporary practice, figures have not only evolved but have become significantly embedded in daily operations. Businesses read through dashboards of candlestick charts and cohort curves (Nison, 2001) that senior management consults in direct preference to prose; engineers coordinate the hundreds of thousands of components that constitute a ship, a hospital or an aircraft through hierarchies of blueprints read without ambiguity by people who may never speak to the designer (Ferguson, 1992); and even life-and-death decisions are made by reading a heartbeat off the grid of an electrocardiogram, or tissue contrast off a radiograph or perfusion map (Goldberger et al., 2017). Music (Treitler, 1984), satellite navigation (Kaplan and Hegarty, 2017), chemistry (Hoffmann, 1995) and network science (Newman, 2010) sustain the same diagrammatic primacy, and every competent empirical paper in the natural and social sciences eventually commits its argument to a figure (Figure 1.1b). **The figure has become the lingua franca of modern knowledge work**, and in many domains the argumentative weight it now carries is strictly greater than the weight carried by the accompanying prose.



(a)



(b)

Figure 1.1: **(a)** A short genealogy of visual argument, from Lascaux and Egyptian wall painting through the medieval *mappa mundi*, Playfair's 1786 bar chart and Snow's 1854 cholera map to modern neural scaling curves. **(b)** The figure as lingua franca of modern knowledge work across business, music, medicine, engineering, navigation, chemistry, networks and the natural sciences. *Sources.* **(a)** Lascaux aurochs, Beni Hasan caravan, Dunhuang star chart, Hereford *mappa mundi*, Playfair 1786 bar chart, Snow 1854 cholera map and Minard 1869 flow map of Napoleon's Russian campaign, all public domain (via Wikimedia Commons, the Metropolitan Museum of Art, the British Library, Hereford Cathedral, the Internet Archive and the Bibliothèque nationale de France respectively); scaling curves adapted from Kaplan et al. 2020. **(b)** A candlestick chart, score excerpt, ECG trace, blueprint, nautical chart, molecular diagram, network graph and scientific plot, from public-domain or openly-licensed repositories (Wikimedia Commons, NOAA, PhysioNet) or rendered by the author.

1.2 Can *Intelligentia Machinae* (Machine Intelligence) Comprehend Visuals Similarly?

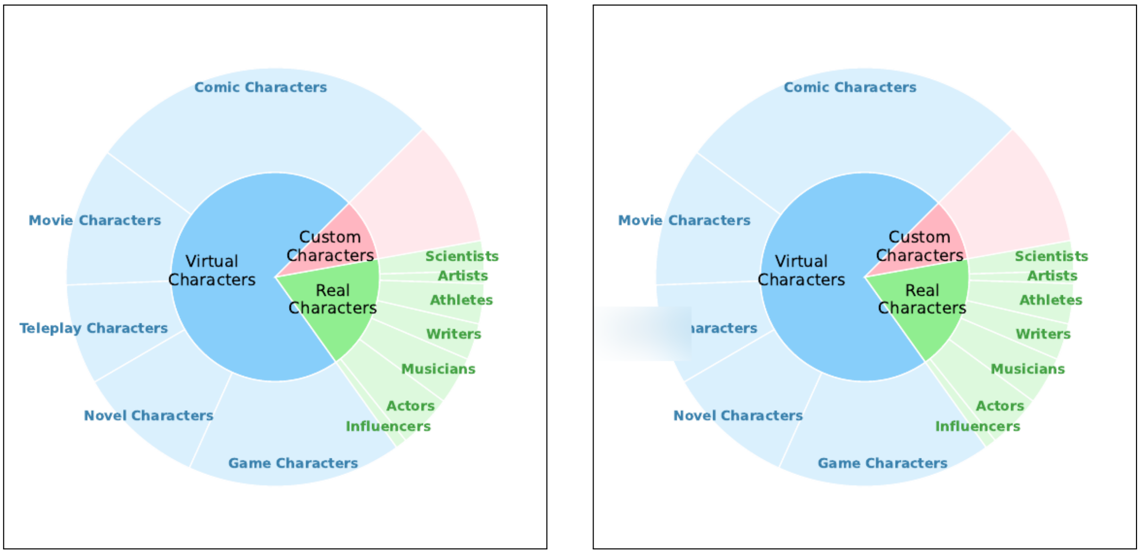
Recent progress in artificial intelligence has been uncommonly rapid, and the current generation of large models has brought within plausible reach an older ambition of the field: a machine that can collaborate with a human being to achieve a shared task. A necessary condition of that collaboration is that the two parties reason on similar planes and navigate near-similar cognitive paradigms. Exact cognitive parity is not required; only a proximity close enough to sustain useful work in concert with a human collaborator. Since the human, as the previous section establishes, is a visual thinker, that proximity obliges the machine to comprehend the visual: to read, interpret and reason from the figure with fidelity comparable to its reading of text.

Systems at the current frontier, including Anthropic's Claude Opus 4.6 and 4.7 (Anthropic, 2026a,b), OpenAI's GPT-5 series (OpenAI, 2025a) and Google DeepMind's Gemini 3 Pro (Google DeepMind, 2026a), are now deployed as infrastructure rather than laboratory curiosities. They are shipped into enterprise software through the major cloud providers, and they are measurably altering how code is written and how commercial operations are conducted. The salient question is therefore no longer whether such systems are capable, but whether that capability extends, in the way the human partner's does, to the visual.

1.3 Motivation for Focusing on Scientific Research Figures

Research is among the first domains in which collaborative intelligence is being tested at scale: between late 2024 and early 2026 the three largest model developers each shipped autonomous research agents (Google, 2024; OpenAI, 2025c; Anthropic, 2025), and scientific deployments such as the Biomni biomedical platform now report research-cycle compressions of an order of magnitude or more (Anthropic, 2026d). Yet scientific knowledge is not monolingual; a substantial share of the record is produced in Chinese, German, Bulgarian and other languages of research, and the cost of excluding that share, in citation, uptake and participation, has been documented at length (Amano et al., 2023). Chart- and figure-understanding benchmarks, however, remain predominantly English: recent multilingual efforts such as PolyChartQA (Li et al., 2025), PM4Bench (Gao et al., 2025) and IndicVisionBench (Sharma et al., 2025) each report substantial cross-lingual degradation on tasks the same models solve comfortably in English. The reach of a multimodal system, considered as a research collaborator, therefore scales with the breadth of languages over which its visual reading remains reliable, and the scientific figure, considered across languages, is the natural object at which to test whether the current frontier reads visually or merely appears to. What that reading must withstand, beyond correctness on the clean image, is illustrated in Figure 1.2.

The failure is not a miscount or a mislabelled axis; it is a confident substitution of a plausible category for evidence the model cannot actually see. Honest reading, not just accurate reading, is therefore the object of interest, and the research questions below are framed accordingly: they ask how faithfully today's frontier models describe scientific figures across languages; how their reading survives visual degradation and adversarial context; and whether, when the evidence is not there, they acknowledge it rather than fill it in.



(a) Original figure. **(b)** One outer label selectively blurred.

Figure 1.2: A sunburst taken from an arXiv paper in the SCIFIG-EVAL corpus. In **(a)**, the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in **(b)**, that single label has been blurred. Asked which category sits between the two, GPT-5.2 answers "*Anime Characters*," with no admission that the relevant region is unreadable. Source image reproduced from arXiv:2505.23923v1 under its open-access licence; blurred variant generated by the SCIFIG-EVAL pipeline.

1.4 Research Questions

The central objective of this thesis is to characterise, across models, languages and adversarial conditions, how faithfully today's frontier multimodal models actually read scientific figures, and to build an evaluation framework that surfaces the failure modes which conventional accuracy benchmarks obscure. Concretely, the investigation is organised around the following five research questions, each pairing a capacity with the conditions under which that capacity is tested:

RQ1 (Description accuracy across languages). To what extent do current vision–language models produce accurate and complete descriptions of scientific figures across typologically diverse languages?

RQ2 (Robustness to visual degradation). How does visual degradation, including noise, compression, aspect distortion and contextual embedding within paper pages, affect the accuracy (whether what is stated is correct), completeness (whether what is visible is reported) and clarity (whether what is reported is expressed unambiguously) of VLM-generated figure descriptions?

RQ3 (Comprehension beyond description). Beyond surface-level description, can VLMs demonstrate genuine comprehension of scientific visualisations, from answering quantitative questions to performing non-trivial inductive reasoning that requires distinguishing analytical insight from surface-level pattern matching?

RQ4 (Adversarial robustness). How robust are VLM outputs when subjected to adversarial manipulations, including misleading captions, contradictory premises and psychologically-informed sycophancy probes?

RQ5 (Calibrated epistemic honesty). When confronted with incomplete or unresolvable visual information, do VLMs exhibit calibrated epistemic honesty, acknowledging uncertainty rather than confabulating plausible but ungrounded responses?

1.5 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 (Background) provides the technical and conceptual foundation: how frontier multimodal models convert figures into internal representations, how scientific figures carry argument rather than merely depict it, and the failure modes and evaluation traditions on which the remainder draws. Chapter 3 (Literature Survey) situates the work against chart understanding, multimodal hallucination, multilingual evaluation and MQM-style scoring. Chapter 4 (Design) sets out the SciFig corpus, the A–R–L framework and the five adversarial probe families. Chapter 5 (Implementation) describes the evaluation pipeline, the model and judge harness and the reproducibility controls. Chapter 6 (Evaluation) reports the main quantitative results across the 11 evaluated models and the four languages, together with ablations on judge ensembles and probe families. Chapter 7 (Discussion) interprets the findings, connects them to the A–R–L framework and acknowledges the limitations of the study. Chapter 8 (Conclusion) summarises the contributions, revisits the research questions and outlines directions for future work.

Chapter 2

Background

This chapter provides technical background on frontier multimodal systems, organised around four questions covered in turn by §2.1–§2.4.

2.1 How do today's Large Language Models see?

Contemporary vision-language models share a three-stage pipeline. An image is partitioned into patches and mapped to a sequence of embeddings by a vision encoder; the embeddings are projected, through a connector, into the token space of a language backbone; and the backbone consumes the projected vision tokens as a prefix alongside the text prompt (Liu et al., 2023; Alayrac et al., 2022; Li et al., 2023a). By the time the image reaches the language model it is a sequence of soft tokens drawn from the same vector space as text, so every downstream operation on the figure is an operation on these tokens.

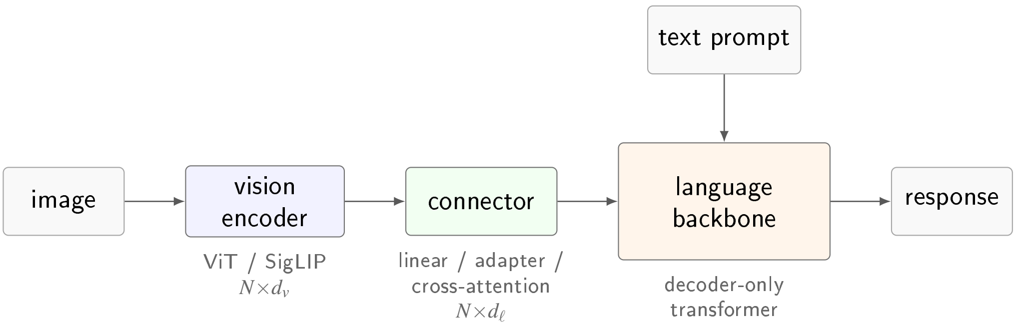


Figure 2.1: The canonical vision-language pipeline. A Vision Transformer (Dosovitskiy et al., 2021), typically trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, maps the image to N patch embeddings of width d_v; a connector (linear projection, adapter, or cross-attention module) retunes them to the backbone width d_ℓ (Liu et al., 2023; Li et al., 2023a); the decoder-only language model consumes them as a prefix alongside the text prompt.

Four architectural choices distinguish frontier families. The *vision encoder* is most commonly a Vision Transformer (Dosovitskiy et al., 2021) trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective; a recent minority, led by Qwen2-VL and Qwen3-VL (Bai et al., 2024; Qwen Team, Alibaba, 2025), train a custom encoder with 2D rotary position embeddings to accept images at arbitrary resolution. The *connector* is a linear projection into the backbone's embedding space (Liu et al., 2023), a cross-attention adapter with learnable queries of the Q-Former or Perceiver variety (Alayrac et al., 2022; Li et al., 2023a), or a set of LoRA modules (Hu et al., 2022) spliced into the backbone's weights as in Phi-4 multimodal (Microsoft Research,

Table 2.1: Publicly documented vision-language architecture across seven representative families spanning the open-weight and closed frontier. "–" denotes a detail that is not disclosed in the primary source. The four open-weight families (top block) publish detailed technical reports; the three closed frontier families (bottom block) publish system cards that address capability and safety but leave the vision pipeline undisclosed. *Sources.* Gemma 3 (Gemma Team, Google DeepMind, 2025); Qwen3-VL (Qwen Team, Alibaba, 2025), with architectural continuity from Qwen2-VL (Bai et al., 2024); Llama 4 (Meta AI, 2025); Phi-4 multimodal (Microsoft Research, 2025); Claude Opus 4.x (Anthropic, 2026c); GPT-5 / 5.2 (OpenAI, 2025b); Gemini 3 Pro (Google DeepMind, 2026b).

Family	Vision encoder	Resolution / tiling	Image tokens	Fusion regime
Gemma 3	frozen SigLIP-400M	896×896 + Pan&Scan	256 (linear proj.)	adapter
Qwen3-VL	custom ViT + 2D-RoPE	dynamic, any aspect	variable (MRoPE)	native
Llama 4	MetaCLIP-derived ViT	–	– (early-fusion)	native early-fusion
Phi-4 mm	CLIP variant + LoRA	–	–	Mixture-of-LoRAs
Claude Opus 4.x	–	$\leq$ 2576 px long edge	–	–
GPT-5 / 5.2	–	–	–	native (claimed)
Gemini 3 Pro	–	up to 900 images / prompt	–	native (sparse MoE)

2025). The *fusion regime* ranges from a late adapter on a frozen backbone, through early fusion on interleaved tokens (Meta AI, 2025; Chameleon Team, 2024), to ground-up native co-training of vision and text from initialisation. The *image-token budget*, finally, varies from a fixed allocation per image (Gemma Team, Google DeepMind, 2025) to aspect-preserving variable-length sequences (Bai et al., 2024), and sets an upper bound on the fine-grained visual detail the encoder can transmit to the backbone. Table 2.1 summarises where seven representative families sit along these axes.

Disclosure along these axes is structurally asymmetric. The open-weight families publish technical reports that document the pipeline end to end, whereas the closed frontier families publish only system cards (Anthropic, 2026c; OpenAI, 2025b; Google DeepMind, 2026b) that leave the encoder, the resolution policy, the token budget and the connector architecture undisclosed. Any evaluation that would attribute an observed output to a specific pipeline stage by reading internal states therefore has, at the frontier, no internal states to read.

2.2 How do today's LLMs reason about what they see, and is it impacted by language?

Reasoning in a vision-language model is autoregressive text generation conditioned on a vision-token prefix. At each decoding step the backbone's self-attention layers attend over a sequence that concatenates the projected vision tokens produced by the encoder with the text prompt, so that every generated token is a function of both channels jointly. Any cognitive operation the model performs on a figure is realised, mechanically, as attention-weighted retrieval from this joint prefix during decoding.

Three prompting regimes are layered on top of this substrate. Under *direct-answer* prompting the model emits its answer immediately; whatever reasoning occurs is implicit in the transformer computation between the prefix and the first output token. Under *chain-of-thought* (CoT) prompting (Wei et al., 2022), the model first emits an explicit natural-language rationale and then the answer; because the rationale is itself generated autoregressively, each of its tokens can re-attend to the vision tokens in the prefix, which provides more decoding steps over which visual evidence can be consulted. Multimodal extensions of CoT augment this loop with image-space scaffolding. Multimodal-CoT (Zhang et al., 2023) interleaves rationale generation with visual features in a two-stage procedure; Set-of-Mark prompting (Yang et al., 2023) overlays enumerated marks on the image so that the rationale can reason over discrete regions by reference; and visual chain-of-thought methods (Shao et al., 2024) condition the model to emit explicit region crops during reasoning, effectively re-querying the encoder on sub-regions as intermediate computation.

The empirical reasoning that emerges from this substrate exhibits a characteristic gap between descriptive and inferential performance. CharXiv (Wang et al., 2024) partitions its evaluation into descriptive and reasoning subsets, and every frontier family scores substantially lower on the reasoning half than on the descriptive half. ChartMuseum (Tang et al., 2025), built from harder naturally occurring charts, reports the same split in sharper form, with the best open-weight model at 38.5%, the best closed model at 63%, and human annotators above 90%. Controlled ablations attribute a large fraction of the reasoning error to the perceptual prefix rather than to the reasoning computation that consumes it. Replacing the raw figure with its ground-truth data table lifts performance sharply on nominally reasoning-heavy questions (Wang et al., 2024; Tang et al., 2025), which indicates that the vision tokens transmitted to the backbone are routinely missing the tick values, axis ranges, and legend associations that the reasoning would otherwise condition on. The image-token budget described in §2.1 places an architectural upper bound on this fidelity, and a growing body of cross-modal analysis shows that the backbone compensates for low-fidelity vision tokens by up-weighting the text channel during attention (Zhang et al., 2023).

Language enters this pipeline at three structurally distinct layers and perturbs reasoning at each. At the *pixel* layer the vision encoder processes non-Latin scripts through the same fixed patch grid it was primarily pretrained on for English; fine-grained symbol recognition (Cyrillic diacritics, Chinese logographs at axis-label size, the tight compounding of German axis labels) degrades before any language-specific reasoning takes place. At the *token* layer the backbone's tokeniser fragments non-English text into finer subword units than English, which lengthens the sequence over which a claim about the figure is represented and dilutes whatever lexical priors the backbone has learned. At the *prior* layer the backbone's pretraining distribution is itself English-dominant, so that for an underdetermined chart the conditional distribution over continuations is systematically biased toward English-plausible content. PM4Bench (Gao et al., 2025) reports that higher-level reasoning accuracy correlates tightly with lower-level OCR accuracy across languages, consistent with the pixel-layer term above dominating the residual error, and PolyChartQA (Li et al., 2025) and IndicVisionBench (Sharma et al., 2025) report pronounced cross-lingual degradation localised at the encoder-OCR stage rather than at the reasoning step that follows.

2.3 How do today's LLMs behave under difficult or adversarial seeing conditions?

Under controlled perturbations of the input, three input-conditions have been studied quantitatively in the recent literature. Each targets a specific stage of the pipeline described in §2.1 and is probed by a dedicated benchmark.

The first condition is visual information loss at the encoder. When regions of the image are blurred, cropped, downsampled, or occluded, the patch grid delivers reduced signal and the projected vision tokens carry diminished content at the affected positions. POPE (Li et al., 2023b) measures the rate at which object-level assertions in the generated output correspond to objects the image actually contains, and reports that frontier VLMs affirm the presence of objects the image does not contain at non-trivial rates on synthetically sampled queries. HallusionBench (Guan et al., 2024) extends the measurement to figure-level visual-reasoning items in which the visual and textual channels are deliberately placed in tension. ChartHal (Xu et al., 2025) isolates the effect on scientific charts through controlled visual ablations in which specified image regions are removed and the model is queried about the removed content; reported abstention rates are low, and the confidence distribution on substituted answers is indistinguishable from that on unperturbed queries. The sunburst in Figure 1.2 of Chapter 1 is an instance of this condition.

The second condition is cross-channel contradiction at the backbone. The backbone's attention weights vision and text tokens jointly, and under controlled manipulation of the text channel the empirical weight placed on vision diminishes. Sharma et al. (2023) formalise this behaviour in pure-text language models as *sycophancy*, the tendency of a model to adopt an interlocutor's implied position when that position is asserted in the prompt, and Zhao et al. (2024) document the visual analogue: VLMs revise their description of a figure in response to leading questions, misleading captions, or asserted prior beliefs, sometimes in direct contradiction of their own earlier output on the same image. A closely related *textual dominance* effect concerns the case in which the prompt asserts a proposition that the image refutes; the asserted proposition is reflected in the output above chance. Zhao et al. (2023) report that small adversarial perturbations injected into either the image or the prompt shift the output distribution by magnitudes comparable to those obtained by replacing the image entirely.

The third condition is inference pressure beyond the information the image supplies. Kadavath et al. (2022) established that text-only language models produce, under suitable elicitation, confidence estimates that correlate with accuracy. Extensions of this protocol to VLMs report that calibration quality decreases with the degree of visual perturbation applied to the input, so that the confidence stated on perturbed inputs does not track the reduced accuracy those inputs induce. Independently, mechanistic analyses tie a fraction of object-hallucination errors to high epistemic uncertainty at specific early vision-encoder tokens and to disproportionate attention on a small number of near-uninformative patches, which relocates some of the error from the reasoning step to the perceptual prefix over which that reasoning operates.

These three conditions define the input axes probed by the behavioural framework developed in Chapter 4, and the output quantities described above are the empirical targets reported in Chapter 6.

2.4 How do we evaluate the seeing proficiency?

Evaluating VLMs on open-ended visual tasks places two technical demands on the measurement apparatus. The first is a rubric more expressive than accuracy on a single-answer question, because the output quantities described in §2.3 do not reduce to a binary correct / incorrect judgement. The second is a grading procedure that does not require a human annotator in the loop for every candidate-prompt pair, since candidate populations produced by modern multilingual and perturbation-based evaluations routinely run to tens of thousands of outputs.

Chart-understanding benchmarks have moved toward this target over three generations. PlotQA (Methani et al., 2020) and ChartQA (Masry et al., 2022) paired largely synthetic charts with short-answer questions amenable to exact-match grading. CharXiv (Wang et al., 2024) and ChartQAPro (Masry et al., 2025) moved to real scientific charts and separated descriptive from reasoning subsets. ChartMuseum (Tang et al., 2025) and ChartHal (Xu et al., 2025) replaced exact-match grading with open-ended description, multi-step reasoning, and controlled abstention, each of which requires an evaluator that can judge free-form output against a reference rubric.

The reference rubric widely adopted for that evaluator is Multidimensional Quality Metrics (MQM), introduced by Lommel et al. (2014) for machine translation and since extended across natural-language generation. Under MQM, each error in an output is tagged with a category from a fixed typology (accuracy, fluency, terminology, style) and a severity weight (minor, major, critical), and a scalar quality score is computed as a weighted aggregation. The apparatus transfers to figure description, where the typology becomes visual fidelity, completeness, and clarity, and the severity weighting distinguishes a misread tick from a fabricated series.

Human MQM annotation does not scale to modern candidate volumes, and the standard alternative is the *LLM-as-judge* paradigm (Zheng et al., 2023), in which a strong general-purpose model grades candidate outputs against a reference rubric. The paradigm is reproducible and, under careful deployment, reaches human-level annotation agreement on several tasks. It also exhibits a documented set of systematic biases. Self-preference, under which a judge awards higher scores to outputs that resemble its own family's style, is reported by Wataoka et al. (2024). Position bias in pairwise comparison is reported by Zheng et al. (2023) and persists at non-trivial magnitude across most judge models. A family of length, verbosity, and concreteness biases rewards more elaborate output largely independent of correctness. Standard mitigations are randomising candidate position, averaging scores across position swaps, and aggregating across a heterogeneous ensemble of judges drawn from different model families. These have been repeatedly validated and constitute the methodological baseline for any open-ended VLM evaluation.

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